

MORUYA HEADS, Australia

WMO: 949370

Lat: 35.909S

Lon: 150.153E

Elev: 55

StdP: 14.67

Time Zone: 10.00 (AUS)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	39.8	41.5	29.1	23.1	56.8	31.3	25.6	55.1	33.2	55.6	30.5	55.2	6.5	250	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.6	80.6	64.7	77.1	67.3	75.3	67.6	71.6	75.8	70.5	74.6	69.4	73.4	15.9	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	110.8	74.3	69.0	106.6	73.3	67.8	102.5	72.3	35.4	76.0	34.5	74.5	33.5	73.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.1	28.4	23.8	DB	36.6	93.4	2.1	5.7	35.1	97.5	33.9	100.8	32.7	104.0	31.2	108.1	(4)
(5)			WB	35.1	73.3	1.5	1.2	34.1	74.2	33.2	74.9	32.3	75.5	31.3	76.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	67.6	68.3	66.4	61.8	58.3	54.7	52.4	53.8	57.0	59.3	62.0	65.0	(6)	
	DBStd	6.52	3.21	3.20	3.48	3.48	3.84	3.53	3.28	4.03	4.60	4.67	4.23	4.08	(7)	
	HDD50	19	0	0	0	0	0	1	12	6	0	0	0	0	(8)	
	HDD65	1998	10	6	22	106	212	309	389	349	247	190	110	48	(9)	
	CDD50	3859	546	513	510	355	257	142	88	123	209	289	361	466	(10)	
	CDD65	364	91	99	67	10	4	0	0	2	6	14	21	50	(11)	
	CDH74	863	155	164	131	25	5	0	0	5	44	127	104	103	(12)	
	CDH80	194	37	16	30	6	0	0	0	0	5	40	32	28	(13)	
(14) Wind	WSAvg	9.8	10.9	10.2	9.7	9.2	8.3	8.5	8.4	9.4	9.9	10.6	11.4	10.8	(14)	
(15) Precipitation	PrecAvg	39.9	3.6	3.9	4.6	3.9	3.1	3.5	2.0	2.5	2.6	3.5	3.6	3.2	(15)	
	PrecMax	69.1	14.3	22.7	22.5	19.9	10.2	20.2	8.8	12.2	7.8	20.7	11.9	9.7	(16)	
	PrecMin	18.4	0.4	0.1	0.2	0.1	0.0	0.0	0.0	0.0	0.1	0.3	0.4	0.4	(17)	
	PrecStd	12.6	2.8	4.8	4.3	4.3	2.6	4.0	2.1	2.9	1.9	3.7	2.6	2.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.3	80.2	80.8	77.2	73.5	69.2	67.4	74.1	80.5	86.2	84.3	83.4	(19)	
		MCWB	64.8	70.4	68.0	65.4	59.2	56.9	54.6	55.5	60.7	63.0	63.8	65.8	(20)	
	2%	DB	77.0	77.2	76.6	73.0	69.5	66.6	63.8	67.4	73.6	76.9	75.9	75.6	(21)	
		MCWB	68.5	70.7	68.6	64.6	60.1	56.4	52.8	54.0	57.8	60.8	63.4	68.4	(22)	
	5%	DB	75.0	75.7	74.4	71.1	67.5	64.2	61.6	64.5	68.9	70.4	71.0	73.5	(23)	
		MCWB	68.4	69.9	67.5	63.6	59.6	55.0	52.3	53.7	57.5	59.2	64.2	67.2	(24)	
	10%	DB	73.6	74.4	72.8	69.4	65.8	62.5	59.8	62.3	65.6	67.0	68.8	71.8	(25)	
		MCWB	67.5	68.9	66.2	62.5	58.6	53.9	51.6	53.9	57.2	59.2	62.9	65.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.3	73.4	72.1	68.4	65.3	60.7	57.6	60.6	63.8	65.5	68.3	71.1	(27)	
		MCDB	77.0	77.2	76.7	73.5	68.4	66.0	62.6	64.5	71.2	75.3	74.1	76.6	(28)	
	2%	WB	70.8	72.1	70.3	66.6	63.7	58.4	55.9	58.5	61.7	63.2	66.5	69.6	(29)	
		MCDB	75.1	75.9	74.5	71.1	67.4	64.3	60.3	63.5	68.2	69.9	72.6	74.2	(30)	
	5%	WB	69.7	70.9	68.7	65.0	61.9	56.8	54.3	56.7	60.1	61.9	65.0	68.1	(31)	
		MCDB	73.5	74.7	73.3	69.7	66.1	62.3	58.8	62.2	66.0	67.9	70.1	72.3	(32)	
	10%	WB	68.5	69.8	67.3	63.5	59.8	55.1	52.9	54.8	58.5	60.6	63.7	66.7	(33)	
		MCDB	72.4	73.6	72.0	68.2	64.1	60.5	58.0	60.8	64.1	66.1	68.0	70.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.8	10.6	11.9	13.8	13.7	14.5	15.2	16.4	15.7	14.4	12.3	11.5	(35)	
		MCDBR	13.4	12.4	14.6	16.0	15.3	16.0	17.2	18.4	21.3	22.1	17.1	14.3	(36)	
	5% WB	MCWBR	7.6	7.6	8.5	9.8	9.6	9.4	10.2	10.2	9.7	9.8	8.3	8.4	(37)	
		MCWBR	11.7	10.9	12.7	14.3	12.5	13.7	14.1	15.9	16.2	15.9	14.2	12.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.422	0.382	0.355	0.328	0.312	0.299	0.290	0.299	0.333	0.350	0.373	0.381	(40)	
		taud	2.349	2.504	2.561	2.610	2.601	2.633	2.654	2.622	2.521	2.504	2.475	2.470	(41)	
	Ebn at Noon		290	295	292	282	269	265	275	288	294	302	303	304	(42)	
		Edh at Noon	42	35	30	25	23	20	21	24	31	34	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2060	1775	1518	1181	915	709	836	1108	1506	1809	1948	2113	(44)	
	RadStd		169	146	111	96	46	53	66	58	102	122	149	140	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.70	N/A	N/A	N/A	N/A	N/A	N/A	-135	+225	+94

Nomenclature: See separate page